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Quantitative motion-tracking methods are widely used in computational linguistics, psychology, and the cognitive sciences to measure fine-grained behavioral dynamics. By contrast, theoretical linguistics—in particular emerging work on gesture and multimodal meaning—has only recently begun to exploit their methodological potential. This paper argues that such tools are particularly well suited to addressing theoretical questions that turn on non-categorical, scalar variation in linguistic expression, where relevant distinctions are encoded in continuous properties of form, as is the case with gesture. As a case study, the paper investigates intensified head nods as gestural response elements in multimodal assertion. When used as standalone responses, head nods can function analogously to affirmative particles such as *yes*, yet their realization varies continuously in kinematic features such as amplitude, velocity, and number of repetitions. The central hypothesis is that this gradient modulation functions as a visual analogue to intensified response particles in speech (e.g., *absolutely*) by strengthening perceived speaker commitment. To test this hypothesis, nod intensity is operationalized through computational kinematic measurement and evaluated in a controlled perception experiment. The results show that increased kinematic intensification systematically correlates with stronger commitment attribution. This paper illustrates how computational approaches can support multimodal measurement in addressing theoretical semantic questions and advancing the empirical study of gesture.

1 Introduction

Recent years have seen growing interest in the formal semantics of gesture and multimodal meaning.¹ Compositional accounts of gesture–speech interaction investigate what gestures contribute to interpretation and how this contribution integrates with spoken content (e.g., Ebert, Ebert, and Hörnig, 2020; Ebert and Ebert, 2014; Schlenker, 2018). At the same time, spatial and vector-based approaches emphasize the visuo-spatial character of iconic gestures and argue for representations that derive semantic structure from aspects of kinematic form (Lücking, Henlein, and Mehler, 2025).

These approaches share the view that gestures are systematic meaning-bearing elements whose contribution can be formally modeled. More broadly, gestures have also been analyzed as forms of demonstration, in which speakers convey information not only by describing events but by performing aspects of them through bodily behavior (Clark and Gerrig, 1990). Unlike lexical items, gestures are temporally unfolding, spatially extended events characterized by continuous variation along multiple dimensions.

Formal analyses necessarily abstract away from much of this internal variation in order to model gesture–speech interaction compositionally. While this abstraction is theoretically indispensable, it leaves open which aspects of gestural realization are meaning-relevant and how they contribute to interpretation. Addressing this question requires methods that render the kinematic structure of gestures empirically accessible and comparable across tokens.

Recent advances in motion tracking and pose estimation make it possible to extract fine-grained movement information from video data. Systems such as MediaPipe (Lugaresi et al., 2019) or OpenPose (Cao et al., 2021) provide high-resolution representations of bodily movement, enabling the quantification of features such as head rotation

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amplitude, velocity, and repetition. Such methods are widely used in psychology and cognitive science to investigate gesture segmentation, temporal organization, and the relation between movement dynamics and communicative structure (e.g., Pouw, Dingemanse, et al., 2021; Pouw, Trujillo, and Dixon, 2020; Trettenbrein and Zaccarella, 2021). Their integration into semantic theory, however, remains limited.

Kinematic analysis is relevant for gesture semantics because it allows measurable properties of physical signal structure to be systematically related to interpretive differences. The present paper argues that such measurement extends the methodological toolkit of multimodal semantics by treating gestures as structured kinematic events whose properties can be associated with interpretive effects.

As a case study, the paper examines intensified head nods as gestural response elements. When used as standalone responses, head nods can function analogously to affirmative particles such as *yes*, yet their realization may vary continuously in amplitude, velocity, and repetition. The guiding hypothesis is that this kinematic modulation functions as a gradient analogue to intensified response particles in speech, such as *absolutely*, by strengthening the perceived degree of speaker commitment.

To test this hypothesis, nod intensity is operationalized through a computational kinematic analysis pipeline and evaluated in a controlled perception experiment. The findings indicate that increased kinematic intensification corresponds to stronger commitment attribution, thereby demonstrating how motion-based measurement can inform semantic analysis.

The broader aim of the paper is to show how computational measurement can contribute to theory building in multimodal linguistics. By operationalizing continuous properties of gestural form, kinematic analysis enables more precise hypotheses about gesture–meaning correspondences and about which aspects of physical signal structure are interpretively active.

The remainder of the paper is organized as follows. Section 2 introduces the theoretical framework required for the case study. Section 3

presents the kinematic operationalization of head nod intensity. Section 4 reports the empirical study testing whether kinematic intensification modulates attributed speaker commitment. Section 5 discusses the broader theoretical and methodological implications of kinematic analysis for gesture semantics, in particular for the relation between physical signal structure and depictive interpretation.

2 Background: Response elements across modalities

This section situates the present case study in two strands of work: (i) formal analyses of response particles in spoken language and (ii) emerging work on gestural response elements. The aim is not to provide a comprehensive survey of response systems, but to isolate a specific configuration central to the present study: cases in which head gestures function as standalone affirmative or rejecting responses, i.e., as independent dialogue moves.

A substantial body of work in theoretical linguistics has investigated response particles such as English *yes* and *no* (e.g., Claus et al., 2017; Farkas and Roelofsen, 2012; Ginzburg and Sag, 2000; Krifka, 2013; Roelofsen and Farkas, 2015).² This literature has largely focused on spoken languages and, within those, on the spoken-auditory modality. Only comparatively recently has attention turned to sign languages (e.g., Loos, Steinbach, and Repp, 2024) and to gestural response elements (e.g., Fritzsche and Walter, 2025; González-Fuente et al., 2015; Li et al., 2016; Loos and Repp, 2025), while formal semantic analyses of visual response elements remain scarce (but see Lücking and Ginzburg, 2021, for an HPSG-based analysis of the head shake).

At first glance, response particles appear straightforward, not least because they are short and frequent items. In particular in response to positive antecedents, their contribution is usually transparent:

²Throughout this paper, I use *response element* as a modality-neutral cover term, encompassing both spoken particles and gestural response forms. The term *response particle* is reserved for lexicalized spoken items.

- (1) a. S₁: Did you feed the cat? / You fed the cat.
 b. S₂: Yes. (= S₂ fed the cat)
 c. S₂: No. (= S₂ did not feed the cat)

With negative antecedents, however, matters are more complex. In English, *yes* and *no* are, in principle, ambiguous and may be used interchangeably with negative antecedents (with speaker- and context-dependent preferences, see e.g., Kramer and Rawlins, 2012):

- (2) a. S₁: Did you not feed the cat? / You did not feed the cat.
 b. S₂: Yes / No (, I did). (= S₂ fed the cat)
 c. S₂: Yes / No (, I didn't). (= S₂ did not feed the cat)

Cross-linguistic research further shows that response systems vary systematically. Languages differ, for instance, in whether they employ truth-based or polarity-based systems (Pope, 1972), and some languages make use of specialized particles for rejecting negative antecedents, such as German *doch* (Blühdorn, 2012; Claus et al., 2017).

Alongside lexical response particles, spoken language interaction frequently employs gestural response elements, most prominently head nods and head shakes. A head nod consists of a vertical movement in the depth dimension, whereas a head shake involves horizontal movement around the vertical axis. Figure 0.1 illustrates these two movement types.

In many Western cultures, head nods are conventionally associated with affirmation and head shakes with rejection. They are commonly classified as emblems, i.e., gestures with culturally conventionalized form-meaning pairings (see e.g., Efron, 1941; Ekman and Friesen,

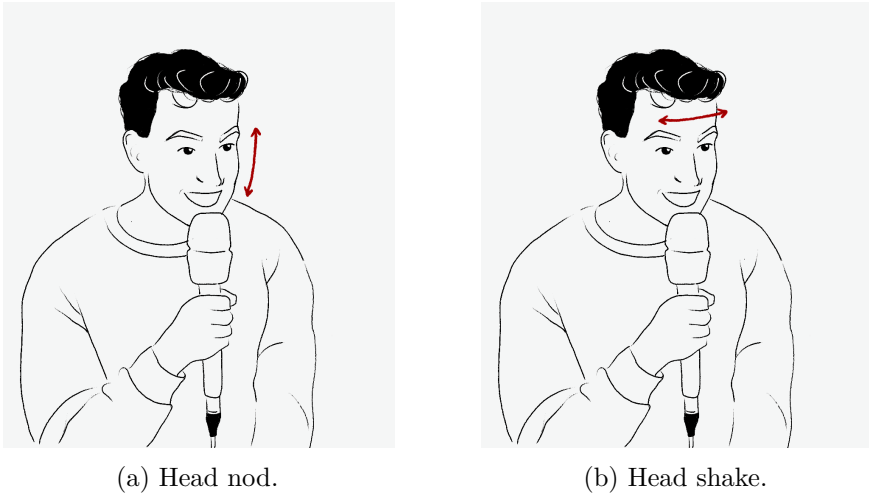


Figure 0.1: Schematic depiction of head nod (a) and head shake (b).

1967).³ At the same time, head movements serve a wide range of communicative functions beyond their use as standalone response elements. They frequently occur as backchannel signals, discourse-structuring cues, deictic markers, or indicators of lexical repair (cf. Kendon, 2002; McClave, 2000). Head nods in particular are among the most common bodily signals in face-to-face interaction and occur extensively in listening contexts (Bauer et al., 2024).

The present study abstracts away from these additional uses and concentrates on cases in which head gestures themselves constitute the response. In such contexts, nods and shakes can substitute for verbal response particles and function as independent assertoric moves:⁴

³See also Ebert and Walter (2025) for a formal approach to emblematic gestures, although their discussion does not explicitly address head gestures.

⁴The equivalence illustrated here represents the simplest configuration, namely usually responses to positive antecedents. In responses to negative antecedents, the mapping between gestural and verbal response forms becomes more complex. For example, experimental evidence suggests that head nods in German can substitute not only for *ja* but also for *doch* in contexts where *ja* would be blocked by the specialized particle *doch* (Fritzsche and Walter, 2025). For

- (3) a. S_1 : Did you feed the cat? / You fed the cat.
 b. S_2 : Yes. / HEADNOD (= S_2 fed the cat)
 c. S_2 : No. / HEADSHAKE (= S_2 did not feed the cat)

Head gestures also frequently co-occur with verbal response particles (González-Fuente et al., 2015; Li et al., 2016; Loos and Repp, 2025). In such multimodal cases, however, the mapping between gestural form and response polarity is not strict: head shakes are common in rejections but can also accompany affirmations, and head nods may also appear in responses that reject a negative antecedent.

In essence, the working assumption adopted here is that, when used as standalone response elements, head nods and head shakes instantiate the same type of assertoric discourse move as the spoken particles they replace. On this view, they are functionally equivalent to verbal response particles at the speech act level.

2.1 The semantics of response particles

This paper adopts an anaphoric approach to response particles, following Krifka (2013) and related work (e.g., Farkas and Roelofsen, 2012; Roelofsen and Farkas, 2015). On this view, response particles are anaphoric to a contextually salient propositional antecedent introduced by the prior move. Crucially, response particles do not merely encode polarity; they realize a speech act, namely an assertion.

The analysis assumes that speech acts are compositionally active at Logical Form (LF) in a dedicated projection (ForceP), headed by covert speech act operators (e.g., Beck, 2016; Krifka, 2014; Sauerland and Yatsushiro, 2017). In particular, the assertion operator ASSERT transforms propositional meanings (type $\langle s, t \rangle$) to context updates (type $\langle c, c \rangle$). A context c minimally contains speaker c_{sp} , hearer c_h , time c_t , and common ground C_w (i.e., the set of worlds compatible

related discussion of the functional differentiation of head shakes from spoken *no*, see Lücking and Ginzburg (2021).

with the assumptions of the interlocutors). The operator is defined as in (4).

$$(4) \quad \llbracket \text{ASSERT} \rrbracket = \lambda p. \lambda c. \iota c' : c' = \langle c_{sp}, c_h, c_t, C_w \cap \{w : \text{assert}(p)(c)\} \rangle$$

The condition $\text{assert}(p)(c)$ holds in a world w iff the speaker publicly takes liability for the truth of p in w . Assertion therefore updates the common ground by adding the speaker's commitment to p .

Within this architecture, Krifka (2013) analyzes response particles as anaphoric speech act operators. Specifically, Krifka assumes the semantics in (5) for *yes* and *no*. In the present study, the same analysis is adopted for the use of HEADNOD and HEADSHAKE when they function as standalone response elements.

- (5) a. *yes* / HEADNOD picks up a salient propositional discourse referent p and asserts it: $\text{ASSERT}(p)$
- b. *no* / HEADSHAKE picks up a salient propositional discourse referent p and asserts its negation: $\text{ASSERT}(\neg p)$
- (adapted from Krifka, 2013, p. 7)

Consider the exchange in (6).

- (6) a. S₁: Is Berlin expensive? / Berlin is expensive.
- b. S₂: Yes. / HEADNOD.
- c. S₂: No. / HEADSHAKE.

S₁'s utterance introduces a propositional antecedent p (= Berlin is expensive). An affirmative response—realized either by *yes* or by a head nod—performs an assertion of p , thereby adding S₂'s commitment to p to the common ground. A rejecting response—realized by *no* or by a head shake—asserts $\neg p$, adding S₂'s commitment to $\neg p$.

On this view, head nods and head shakes do not differ in their core semantic contribution from their spoken counterparts. They instantiate the same type of assertoric move. As the next subsection argues, response moves may also vary in *strength*, understood as the degree of speaker commitment.

2.2 Intensified response particles

A growing body of work investigates *intensified response particles* in spoken language: response forms that, beyond performing a basic affirmative or negative move, convey heightened endorsement or rejection, typically understood as a stronger-than-default degree of speaker commitment (e.g., Gotzner, 2022; Greenberg and Wolf, 2019). In English, this family includes affirmative forms such as in (7).

- (7) a. S₁: John is home. / Is John home?
 b. S₂: Sure! / Absolutely! / Of course! / No question about it!
 c. S₂: No way! / Hell no!

(Greenberg and Wolf, 2019, p. 61)

A central claim in this literature is that intensified particles operate at the speech act level. Greenberg and Wolf (2019) analyze Hebrew *legamrey* ('absolutely'), illustrated in (8), as a degree modifier of a gradable speech act operator, specifically ASSERT. Crucially, the relevant scale in their analysis is not one of commitment but of epistemic strength or credence: the particle retrieves a salient antecedent proposition p_c and indicates that the speaker's credence in p_c is maximal. Still, the formal implementation with a degree argument can be transferred rather naturally to a commitment-based analysis, as already noted in Greenberg (2017) and assumed here.

- (8) a. S₁: ha-agartal male/yakar/nafal.
 ('The vase is full/expensive/fell down.')

b. S₂: *legamrey*!

(Greenberg and Wolf, 2019, p. 63)

Formally, *legamrey* combines with a gradable speech act operator G (here: ASSERT), an anaphoric proposition p_c , a degree d , and a context c , forcing d to equal the maximal value on the contextually relevant scale S_G :

- (9) $\llbracket \textit{legamrey} \rrbracket = \lambda G. \lambda p_c. \lambda d. \lambda c. \iota c'. c' = \langle c_{sp}, c_h, c_t, C_w \cap \{w : d = \max(S_G) \wedge G(p_c)(d)(c)\} \rangle$

(Greenberg and Wolf, 2019, p. 65)

Similarly, Gotzner (2022) argues that German colloquial *voll* (lit. 'full', roughly 'totally') has a response particle use in which it contributes a high—though not necessarily maximal—degree of commitment:

- (10) a. S₁: Manfred ist ein Genie. ('Manfred is a genius.')

b. S₂: Voll! ('Totally!')

(Gotzner, 2022, p. 28)

Taken together, these analyses support a view according to which intensified response particles modify the degree argument of a gradable speech act operator. Commitment is thus treated as a scalar dimension associated with assertion, and intensified particles fix this degree at a high or maximal value.

This view has direct consequences for multimodal response systems. If head gestures, when used as standalone response elements, instantiate the same assertoric operator as spoken particles (Section 2.1), and if assertion is associated with a gradable commitment parameter, then

variation in gestural realization may likewise encode differences in commitment strength.

Unlike spoken intensified particles, head gestures do not lexicalize degree. Any modulation must therefore be realized through systematic variation in the physical signal itself. On this hypothesis, kinematic differences in, for example, amplitude, velocity, or repetition function as a non-lexical analogue of degree modification at the speech act level.

2.3 Kinematic modulation of head movements

Independent evidence that head movement form can encode functional distinctions comes from sign language research, where phonetic variation in head movements is well established as functionally relevant (e.g., Bauer et al., 2024; Puupponen et al., 2015; Wilbur, 2000). In languages of the visual-gestural modality, head movements form part of a structured articulatory system, and fine-grained variation in their realization is known to correlate with semantic and pragmatic differences.

For American Sign Language (ASL), Wilbur (2000) (see also Liddell, 1986) reports that large and deliberate nodding serves the function of strong assertion. According to Wilbur (2000, p. 230), “a repetitive head nod conveys what semanticists refer to as the speaker’s commitment to the truth of the assertion”. In Finnish Sign Language (FinSL), Puupponen et al. (2015) likewise associate nodding movements with strongly affirmative functions: repeated nods tend to signal strong affirmation, while a single nod typically signals a simpler affirmative function, though single nods can also convey strong affirmation.

More generally, sign language work points to a set of phonetic or kinematic parameters that differentiate head movement functions, including amplitude, velocity, direction of movement, and number of repetitions (cf. Bauer et al., 2024). In their data on affirmative responses in German Sign Language (DGS), Bauer et al. (2024) find substantial variability and show that affirmative nods differ systemati-

cally from feedback nods: they tend to be larger in amplitude, shorter in duration, and faster in their production.

Taken together, these observations provide empirical support for the idea that head gestures can be systematically modulated to strengthen an assertive move. Crucially, the relevant distinctions are encoded in measurable kinematic parameters rather than in lexical contrasts. Combined with a commitment-based semantics of response elements, this yields a clear prediction: if head gestures can serve as response elements, then continuous variation in their kinematic realization (e.g., amplitude, peak velocity, number of repetitions) should correlate with differences in attributed speaker commitment.

The remainder of this paper tests this prediction by defining a transparent kinematic intensity metric for head nods and using it to construct stimulus contrasts in a controlled perception experiment on commitment attribution.

3 Kinematic operationalization of gesture intensity

This section introduces the methodological core of the paper: a procedure for quantifying the intensity of head nod responses, with potential applicability to gesture sequences more generally. The aim is to operationalize continuous properties of gestural form in a way that renders the hypothesized scalar modulation of assertion empirically testable.

If intensified head nods function as a gradient analogue to intensified response particles, their strengthening effect must be realized in the physical signal itself. For head gestures, the most plausible locus of intensification is systematic variation in kinematic parameters. A principled measure of nod intensity must therefore be grounded in observable movement properties while remaining sufficiently transparent and interpretable for experimental implementation.

The approach taken here treats a head nod as a structured kinematic configuration. A nod is not a single undifferentiated movement, but a configuration defined along multiple measurable dimensions,

for example, its amplitude (the extent of vertical displacement), its temporal dynamics (how fast it moves), and its repetition structure (how many nod cycles occur within the gesture).

Computational kinematic analysis renders these dimensions measurable and explicit by representing the gesture as a continuous time series of head rotation values. This time-resolved kinematic trace provides a quantitative representation of the gesture’s physical realization. From this signal, distinct features—such as peak count, peak prominence, and peak frequency—can be extracted. The intensity metric proposed below is defined over these measured features.

For any given head gesture sequence, here, *intensity* is defined as a composite measure of three kinematic parameters. Drawing on sign language research showing that affirmative nods tend to involve greater amplitude and higher velocity than feedback nods (Bauer et al., 2024), the following features are used:⁵

- (11) a. *Number of nod peaks*: the total number of discrete nod peaks within the gesture sequence.
- b. *Average prominence*: the mean amplitude of nod peaks across the gesture sequence.
- c. *Peak frequency*: the number of nod peaks per second (Hz), reflecting gesture velocity.

Figure 0.2 schematically illustrates these measures on an amplitude–time plot. Each local extremum in the stroke direction is treated as a peak. Variation in peak height (amplitude), peak spacing (frequency), and total peak count jointly contributes to overall intensity.

⁵The feature set is not intended to be exhaustive. In principle, additional kinematic parameters (e.g., angular velocity, acceleration, total duration, or three-dimensional trajectory properties) as well as co-occurring facial features (e.g., brow raise amplitude) could be incorporated into a higher-dimensional model. The present study restricts attention to three features to ensure transparency and interpretability of the intensity metric.

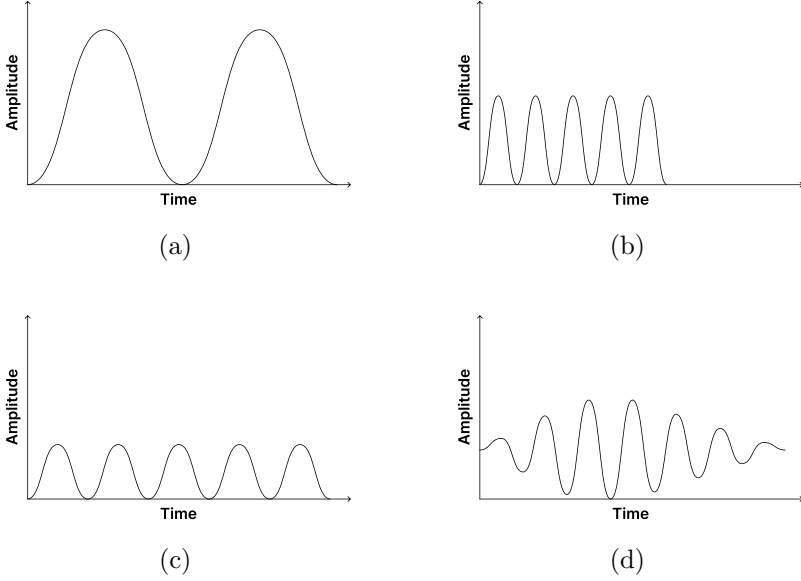


Figure 0.2: Schematic illustration of the kinematic parameters used to quantify nod intensity on an amplitude–time plot. Across panels, peak height (amplitude), peak spacing (frequency), and number of peaks vary. Curves are illustrative and not to scale.

For each gesture sequence, the three measures in (11) are extracted from the kinematic analysis. To ensure comparability across tokens, each measure is normalized to the 0–1 range by dividing it by the maximum value observed across all extracted nod sequences in a given recorded sample. Max-normalization was chosen to preserve proportional differences between tokens while maintaining interpretability on a bounded 0–1 scale. The normalized values are then averaged to yield a single composite intensity score, cf. (12).

$$(12) \quad \text{Intensity} = \frac{1}{3} \left(\frac{\text{n_peaks}}{\max(\text{n_peaks})} + \frac{\text{avg_prominence}}{\max(\text{avg_prominence})} + \frac{\text{peak_frequency}}{\max(\text{peak_frequency})} \right)$$

For interpretability in experimental design and reporting, intensity values may additionally be binned into three equally sized categories, see Table 0.1.

Normalized intensity range	Category
0.00 – 0.33	Weak
0.34 – 0.66	Moderate
0.67 – 1.00	Strong

Table 0.1: Binning of the composite normalized intensity score into three equally sized categories.

To illustrate the calculation, consider a head-nod sequence with the following raw values in (13).

$$\begin{aligned}
 (13) \quad & \text{number of peaks} = 9.0 \\
 & \text{average prominence} = 7.57 \\
 & \text{peak frequency} = 2.63
 \end{aligned}$$

Assume that the maximum observed values across the full set of recorded nod sequences are:

$$\begin{aligned}
 (14) \quad & \max(\text{n_peaks}) = 9.0 \\
 & \max(\text{avg_prominence}) = 13.96 \\
 & \max(\text{peak_frequency}) = 2.63
 \end{aligned}$$

The normalized composite score is then:

$$(15) \quad \text{Intensity}_{\text{example}} = \frac{1}{3} \left(\frac{9.0}{9.0} + \frac{7.57}{13.96} + \frac{2.63}{2.63} \right) = \frac{1}{3}(1 + 0.54 + 1) \approx 0.85$$

This gesture thus receives a composite intensity score of approximately 0.85 and would be classified as *strong* on the normalized scale.

4 Case study: Kinematic intensity and commitment attribution

This experiment investigated whether the intensity of head nods modulates attributed speaker commitment. Two verbal responses as baseline (*ja* ‘yes’, *voll* ‘totally’) and four gestural conditions varying in intensity (HEADNOD-weak, HEADNOD-moderate, HEADNOD-strong, HEADNOD-strong + PALM-UP) were tested.

The additional HEADNOD-strong + PALM-UP condition was included because palm-up gestures recurrently co-occurred with high-intensity nods in the dataset used for the study. Rather than treating these co-occurrences as incidental, a separate condition was included to test whether multi-channel reinforcement systematically modulates commitment attribution beyond head movement intensity alone. While the present study treats the palm-up gesture primarily as an additional manual cue potentially reinforcing commitment, palm-up gestures have also been argued to serve broader discourse-pragmatic functions, for instance signaling shared knowledge or epistemic stance in interaction (e.g., Laparle, 2022). Gesture research further suggests that palm-up gestures belong to a broader family of recurrent communicative actions with related interactional functions (Müller, 2004). The present design does not attempt to adjudicate between these interpretations, but allows us to test whether the addition of such a manual gesture modulates perceived commitment relative to head movement alone.

On the basis of the theoretical considerations outlined above, the following research questions were formulated:

- (16) a. **RQ₁**: Does perceived commitment increase monotonically with head nod intensity?
- b. **RQ₂**: Do manual and non-manual gestures combine to increase perceived commitment?

- c. **RQ₃**: How do commitments to (intensified) gestural responses compare to commitments to (intensified) verbal responses?

4.1 Method

Participants

A total of 1,500 native speakers of German participated in the study. Participants were recruited online via Prolific and provided informed consent prior to participation. The age range was 21–66 years ($M = 36.8$, $SD = 10.6$). 58% of participants identified as female, 39% as male, and 3% as non-binary. All participants self-reported German as their first language and were naive to the purpose of the study.

Computational pipeline and feature extraction

The gestural stimuli used in the experiment were drawn from three episodes (130.45 minutes) of a German video podcast featuring dyadic face-to-face interaction in a videoconference format. The frontal camera perspective enabled frame-wise head-pose tracking.⁶

A total of 241 standalone head nods produced by the same speaker were extracted from the recordings. To assign intensity values to these tokens, all nods were subjected to the kinematic feature extraction analysis.

The input to this procedure consists of a continuous time series of head rotation angles sampled frame by frame from video, from which peak-based features (peak count, peak prominence, and peak frequency) were extracted.

Kinematic tracking was performed with Google’s MediaPipe Holistic model (Lugaresi et al., 2019), which provides dense face-mesh

⁶Analysis scripts, stimulus metadata, and supplementary materials are publicly available via the Open Science Framework at <https://osf.io/b459j/>.

landmarks (478 points) alongside body and hand keypoints. Since MediaPipe does not output head rotation angles directly, head orientation was estimated from selected face-mesh landmarks using the *Face Tracking with Rotations* module implemented in Pouw (2024), building on Thierfelder (2024). This procedure yields frame-wise rotation angles (in degrees) around the three canonical axes (pitch, yaw, roll). For the present analysis, pitch served as the primary signal for head nodding.

For each nod token, the analysis returned a pitch trajectory indexed by video frame. Prior to feature extraction, trajectories were smoothed to reduce frame-to-frame jitter using a Savitzky–Golay filter (window length w , polynomial order k), following the ViCom analysis workflow (Pouw, 2024). Trajectories were additionally mean-centered within token to remove constant offsets due to camera angle and resting head posture.

Peak-based features were then computed on the pitch time series. Local maxima and minima were detected using a standard peak-finding algorithm (`scipy.signal.find_peaks`), subject to a minimum inter-peak distance and a minimum prominence criterion to exclude noise-related micro-oscillations. The total number of detected extrema provides the *peak count*. Peak *prominence* was computed for each extremum (`scipy.signal.peak_prominences`) and averaged across extrema to obtain *average prominence*. Finally, *peak frequency* was calculated as the number of detected extrema divided by token duration (in seconds), yielding a tempo-like measure of nod density.

All scripts follow the envisionBOX workflow and are documented in the public tutorials for *Face Tracking with Rotations*⁷ and the *ViCom*⁸ feature-analysis pipeline.

⁷https://envisionbox.org/embedded_face_tracking_rotations.html

⁸https://www.envisionbox.org/embedded_vicom_analysis_pipeline.html

Materials and design

The experiment employed a single discourse context adapted from an attested exchange in the podcast. Participants read the written context in (17).

- (17) Petra und Dario unterhalten sich über Darios neue Wohnung und überlegen, in welcher Farbe er sein Arbeitszimmer streichen könnte. Petra sagt: “Mit Weiß kann man nix falsch machen!”
 (‘Petra and Dario are discussing what color Dario should paint his new study. Petra says: You can’t go wrong with white!’)

The antecedent assertion was identical across all conditions and was presented in written form. The response by Dario was presented as a short video clip. Only the response element varied.

The design was single-item and one-factorial, with the factor RESPONSE ELEMENT comprising six levels:

- (18) a. ja (‘yes’)
 b. voll (‘totally’)
 c. HEADNOD-weak [Intensity = 0.24]
 d. HEADNOD-moderate [Intensity = 0.67]
 e. HEADNOD-strong [Intensity = 0.87]
 f. HEADNOD-strong + PALM-UP [Intensity = 0.85]

The two verbal conditions served as interpretive anchors. *Ja* represents the default affirmative response particle in German, whereas *voll* is used as an intensified response particle (Section 2.2), thereby spanning distinct points along an intuitively perceived commitment scale.

From the 241 analyzed nod tokens, three were selected to represent distinct regions of the empirically observed intensity distribution

(low, medium, high), ensuring that the experimental manipulation reflects naturally occurring scalar variation in the speaker’s gestural behavior. The additional HEADNOD-strong + PALM-UP condition was included because palm-up gestures recurrently co-occurred with high-intensity nods in the recorded material, allowing us to explore whether multimodal reinforcement further modulates commitment attribution.

Crucially, all gestural stimuli functioned as standalone response elements without simultaneous speech, ensuring that any observed differences in interpretation can be attributed to gestural form rather than verbal content.

Figure 0.3 provides schematic illustrations of the four gestural stimulus conditions. Panel (a) depicts the weak head nod with the lowest intensity value, panel (b) shows the moderate nod, panel (c) shows the strong nod, and panel (d) shows the strong nod accompanied by a palm-up gesture, included here to illustrate the orchestration of two distinct gestural components within a single response.

Procedure

The study employed a single-trial between-subjects design. Each participant saw the written context in (17) paired with one response clip corresponding to one level of the RESPONSE ELEMENT factor.

The experiment was conducted online using SoSci Survey (Leiner, 2024). Participants first read the context and watched the response video on the same page. They could replay the video as often as they wished before proceeding.

Commitment attribution was measured via a deniability-based diagnostic (following Weissman, 2024). Rather than directly asking how committed the speaker was, participants evaluated the acceptability of a subsequent denial. The logic of this diagnostic is that publicly asserting a proposition creates a commitment that cannot easily be disavowed. If a denial is judged acceptable, commitment is perceived

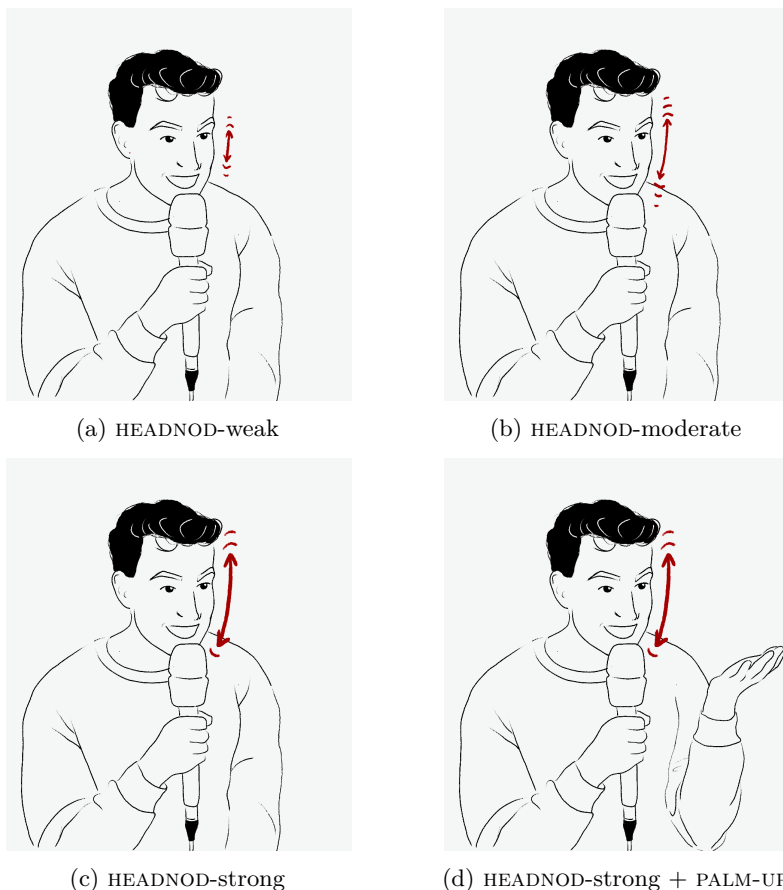


Figure 0.3: Schematic gestural stimulus conditions used in the experiment. Red movement annotations depict gesture intensity; stronger markings indicate higher values on the intensity scale.

as weak; if it is judged unacceptable, commitment is perceived as strong.

Participants rated the denial statement in (19) on a 7-point Likert scale (1 = totally disagree, 7 = totally agree).

- (19) Wenn Mirella später hört, dass Dario sagt, Weiß sei eine furchtbare Wandfarbe, und ihn daraufhin damit konfrontiert, dass er ihrer Empfehlung zugestimmt habe, könnte Dario den Vorwurf überzeugend mit folgendem Satz zurückweisen: “Ich habe nie zugestimmt, dass Weiß eine gute Wandfarbe ist.”
- (‘If Petra later hears Dario say that white is a terrible wall color and confronts him about having agreed with her suggestion, Dario could convincingly dismiss the accusation with the following response: I never agreed that white is a good wall color.’)

Higher ratings indicate greater perceived deniability and thus lower attributed commitment. Lower ratings indicate reduced deniability and therefore stronger attributed commitment.

Hypotheses and predictions

Based on the theoretical considerations above, the following hypotheses were formulated:

- (20) a. **H₁**: Commitment increases with head nod intensity.
- b. **H₂**: Adding a palm-up gesture increases attributed commitment relative to a strong head nod alone.
- c. **H₃**: The intensified verbal response *voll* yields higher attributed commitment than the other conditions.

Taken together, these hypotheses predict a main effect of RESPONSE ELEMENT on deniability ratings.

4.2 Results

Figure 0.4 displays mean deniability ratings (lower ratings indicate stronger perceived commitment) by condition.

The cumulative link model revealed a significant main effect of RESPONSE ELEMENT on deniability ratings ($\chi^2(5) = 1133.4$, $p < .001$).

Tukey-adjusted pairwise contrasts (Table 0.2) showed that the intensified verbal response *voll* yielded significantly lower deniability ratings than every other condition (all $p < .001$), indicating the strongest attributed commitment overall.

The neutral verbal response *ja* did not differ from HEADNOD-strong ($p = .593$), but yielded higher deniability than HEADNOD-strong+PALMUP ($p < .001$) and lower deniability than HEADNOD-moderate and HEADNOD-weak (all $p < .001$).

Within the gestural conditions, a graded pattern emerged: HEADNOD-strong+PALMUP < HEADNOD-strong ($p = .035$), HEADNOD-strong < HEADNOD-moderate ($p < .001$), and HEADNOD-moderate < HEADNOD-weak ($p < .001$).

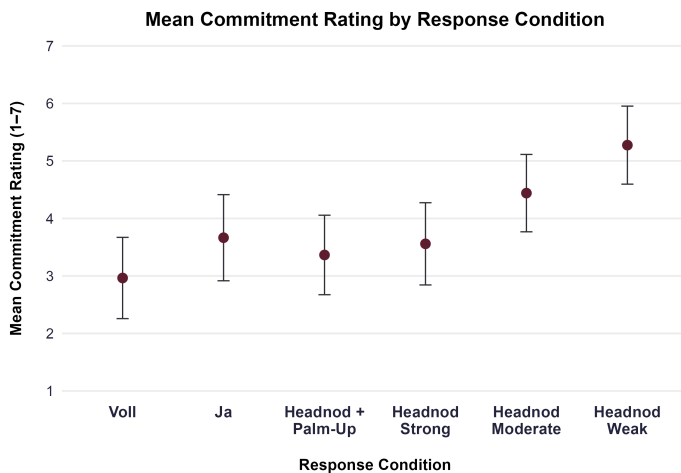


Figure 0.4: Mean commitment-deniability ratings by response condition. Lower ratings indicate stronger perceived commitment. Error bars represent standard deviations.

Contrast	Estimate	SE	<i>p</i> (Tukey)
Voll – Ja	−1.673	0.162	< .001
Voll – HN Strong+PalmUp	−0.957	0.157	< .001
Voll – HN Strong	−1.421	0.161	< .001
Voll – HN Moderate	−3.597	0.178	< .001
Voll – HN Weak	−5.819	0.209	< .001
Ja – HN Strong+PalmUp	0.716	0.156	< .001
Ja – HN Strong	0.253	0.157	= .593
Ja – HN Moderate	−1.924	0.166	< .001
Ja – HN Weak	−4.145	0.197	< .001
HN Strong+PalmUp – HN Strong	−0.464	0.156	= .035
HN Strong+PalmUp – HN Moderate	−2.640	0.170	< .001
HN Strong+PalmUp – HN Weak	−4.861	0.201	< .001
HN Strong – HN Moderate	−2.176	0.167	< .001
HN Strong – HN Weak	−4.398	0.199	< .001
HN Moderate – HN Weak	−2.221	0.177	< .001

Table 0.2: Tukey-adjusted pairwise contrasts from a cumulative link model (ordinal logit) of deniability ratings by condition. Estimates are differences on the latent logit scale (first minus second). Negative values indicate lower deniability (higher commitment) for the first condition.

4.3 Discussion

The study tested whether kinematically measurable variation in head nod form systematically modulates attributed speaker commitment, and how such gestural responses compare to verbal baselines.

Three main findings emerge from the experiment.

First, within the purely gestural conditions, commitment increased monotonically with nod intensity (weak < moderate < strong).

Second, adding a palm-up gesture to a strong nod further increased attributed commitment. This suggests that multimodal cues can

combine cumulatively, supporting a compositional view of intensification across channels. Visible signals from different articulators (head, hand) appear to jointly strengthen the perceived force of the assertion. Third, the intensified verbal particle *voll* yielded the strongest commitment overall. However, a sufficiently strong gestural response approached—and in the multimodal condition even exceeded—the commitment attributed to a neutral verbal assent (*ja*). This shows that gesture is not categorically weaker than speech in conveying commitment. Rather, both modalities appear to operate on a shared scalar dimension of assertion strength. It is worth noting that even the strongest conditions yielded relatively moderate values on the deniability scale. Participants rarely selected the lowest points of the scale, which would correspond to maximal attributed commitment. This may reflect a general reluctance to attribute very strong commitment on the basis of a single response element. Importantly, however, the systematic ordering across conditions indicates that participants were highly sensitive to the gradient modulation of gestural intensity.

5 General discussion: Kinematic structure and depictive interpretation

The results of the experiment show that systematically measured variation in head nod realization predicts systematic variation in attributed speaker commitment. Participants reliably mapped differences in amplitude, repetition, and temporal density onto differences in assertion strength. This supports a gradable conception of commitment at the speech act level and demonstrates that scalar properties of gestural form can modulate discourse interpretation in a principled way.

Beyond these specific findings, the study bears on a broader theoretical issue in gesture semantics: how depictive meaning relates to the structure of physical signal realization. Recent work distinguishes sharply between descriptive symbolic content and depictive iconic content (e.g., Davidson, 2024). On this view, symbolic expressions contribute propositional structure: their meaning is encoded in abstract

representations that compose and license reasoning over alternatives. Depictions, in contrast, do not contribute propositional content in the same way. Instead, they introduce what Davidson calls *particulars*—concrete representational tokens whose interpretation proceeds via perceptual or structural mapping rather than lexical abstraction.

This perspective aligns closely with earlier work on demonstration in communication. Clark and Gerrig (1990) argue that speakers frequently convey information not only by describing events but by *demonstrating* aspects of them through performance. Demonstrations are complex signals in which different aspects of the produced behavior serve different communicative roles. Some aspects of the signal are intended to be *depictive*, representing properties of the referent, while others are *supportive*, enabling the production of the demonstration. Additional aspects may be *annotative*, providing commentary on the depicted content, while remaining aspects are incidental to the motor execution of the behavior. Successful interpretation therefore requires recipients to recognize which aspects of the produced signal are intended to function as representations of the relevant situation.

If gestures function as demonstrations in this sense, their interpretive contribution cannot depend on the gesture ‘as a whole’. Gestures are temporally unfolding motor events that vary along multiple continuous dimensions, including amplitude, velocity, trajectory, timing, and repetition. If depictive meaning arises through structural mapping between aspects of the signal and aspects of the represented situation, then interpretation must involve the selective uptake of particular dimensions of this physical realization. Identifying which dimensions are interpreted as depictive thus becomes a central empirical question for gesture semantics.

At this point a methodological challenge arises. Gesture signals are high-dimensional and continuously variable, yet only some aspects of their realization appear to contribute systematically to interpretation. Without decomposing gestures into analyzable features of movement, depiction risks functioning as a global label rather than a feature-sensitive semantic mechanism. Although theories of demonstration distinguish between depictive, supportive, and annotative aspects of

performance, they typically do not specify how these distinctions can be identified at the level of the physical signal.

One theoretical perspective that helps clarify this issue comes from the structure-mapping framework developed in cognitive science (Emmorey, 2014; Gentner, 1983). In this framework, interpretation is not assumed to rely on global similarity between two representations, but on systematic correspondences between structured relations in a source and a target representation. Crucially, structure-mapping theories predict that interpreters preferentially align relational structure rather than isolated attributes, and that differences are most salient when they are *alignable*, that is, when they correspond to comparable elements within a shared relational structure. Applied to gesture interpretation, this perspective suggests that not every kinematic property of a gesture will be equally relevant for meaning construction. Instead, observers are likely to interpret those aspects of the signal that can be systematically aligned with dimensions of the represented event or communicative act, while other aspects of the movement remain supportive or incidental. In this sense, the challenge of identifying depictive structure in gesture can be reframed as the task of determining which dimensions of physical realization constitute such alignable correspondences. Computational kinematic analysis provides a natural methodological tool for this task, because it decomposes gestures into independently measurable features and thereby makes it possible to test which of these dimensions are consistently mapped onto interpretive differences.

In the present case, peak count, prominence, and temporal density were shown to correlate with attributed speaker commitment. This suggests that these particular dimensions of movement are candidates for depictive structure in the sense of Clark and Gerrig: they are aspects of the signal that interlocutors interpret as representing properties of the communicative act.

More generally, kinematic feature analysis opens a path toward empirically distinguishing different functional components within gestural demonstrations. Features that reliably influence interpretation can be treated as candidates for depictive dimensions. Other aspects of

the movement may instead function as supportive components of the motor performance, enabling the realization of depictive structure without themselves contributing meaning. Still others may operate as annotative features, providing commentary or stance toward the depicted content. Remaining aspects of the signal may be incidental to the motor execution of the gesture.

From this perspective, the contribution of the present study extends beyond the specific notion of commitment. The experiment illustrates a more general methodological strategy for gesture theory: if depictive meaning depends on selective mapping between dimensions of a physical signal and dimensions of interpretation, then identifying those dimensions becomes both an empirical and a theoretical task. Kinematic analysis provides the tools necessary to pursue this task in a systematic and experimentally tractable way.

The broader claim of this paper is therefore that computational kinematic analysis supplies a crucial interface between continuous motor execution and formal semantic theory. For approaches that treat depiction and demonstration as genuine components of meaning, such an interface is indispensable. Without articulating which dimensions of gesture are meaning-bearing, the link between physical signal structure and semantic interpretation remains underspecified. With it, depiction can be investigated as a structured, feature-sensitive mechanism within a formally explicit theory of multimodal meaning.

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